Proposal

The impact of Working From Home on the Real Estate Markets

Question:

How does Working From Home impact on Real Estate Markets in California ?

Motivation

Remote work has become a controversial topic worldwide. In recent years, companies such as Google, Tesla, and Apple have increasingly pushed employees to return to in-person work arrangements. However, many workers prefer remote work, as it reduces commuting costs and increases flexibility.

As a result, employees who are able to work from home may be more likely to remain in their current locations or choose to purchase homes farther from city centers, since daily commuting is no longer a major concern. This shift in residential preferences is expected to change housing demand and lead to fluctuations in housing prices across California. A key motivation of this study is to apply economic models to identify the causal effects of working from home on real estate markets.

Expected Contribution

This study aims to make several contributions to the literature on labor markets and urban economics.

Causal Evidence

- This study provides credible evidence estimating the magnitude of the impact of working-from-home adoption on housing prices and rental markets.

Policy Implications

- The findings can help policymakers better plan for long-run economic development by informing decisions related to infrastructure investment, transportation planning, and housing policy in response to changing residential demand.

Methodological Contribution

- By applying Synthetic Control methods and fixed-effects models, this study demonstrates robust identification strategies for evaluating large-scale labor market shocks and their effects on real estate markets.

Preliminary literature context

Many previous studies have explored working from home and its effects across various fields. The first paper, Why Working from Home Will Stick (2021), illustrates that the adoption of working from home reduces commuting costs while maintaining productivity in many occupations. The authors argue that this shift in work arrangements is likely to be permanent rather than temporary, extending beyond the pandemic period. This study provides evidence that remote work influences where people choose to live.

The second paper, The Donut Effect of COVID-19 on Cities (2022), shows that U.S. cities with a higher proportion of workers able to work from home experienced population outflows and weaker housing demand in urban cores. In contrast, housing demand increased in suburban and rural areas. These findings suggest that working from home significantly reshapes housing demand across regions.

Overall, the existing literature provides strong support for further research on the impact of working from home on real estate markets.

Data Sources:

This research combines multiple datasets that are publicly available from government agencies and well-established data providers.

Housing Prices

- Zillow Home Value Index (ZHVI) and Zillow Observed Rent Index (ZORI) at the county level
- Time period: 2018–2026

Working-from-Home Measure

- American Community Survey (ACS): proportion of workers working from home by county

Other Control Variables

- Federal Reserve: interest rates over time
- American Community Survey (ACS): income and employment measures

These datasets are well structured and available at annual or monthly frequency, making them suitable for panel-data methods such as Difference-in-Differences and Synthetic Control. The sample period covers both pre- and post-2020 years, allowing for clear comparisons before and after the remote-work shock.

Initial empirical strategy idea

The primary empirical strategy is a Difference-in-Differences (DiD) approach that exploits variation in remote-work exposure across counties in California. Counties with a high share of workers working from home are classified as the treatment group, while counties with lower exposure serve as the control group.

The main outcome variables are housing prices and rental prices. The regression specification includes county fixed effects to control for time-invariant local characteristics, as well as time-varying control variables such as interest rates, income, and population.

To strengthen identification, the study also applies a Synthetic Control method. A county with a high density of remote work is selected as the treated unit, and a counterfactual control group is constructed from counties with lower remote-work exposure. This approach improves pre-treatment fit, helps address potential violations of the parallel-trends assumption, and provides more credible estimates of the impact of remote work on real estate markets.

Anticipated Challenges:

To estimate the causal effects of working from home, this study faces several important challenges:

Macroeconomic Shocks

- The expansion of working from home coincided with other major shocks, such as historically low interest rates and large migration flows. These concurrent events make it difficult to isolate the effect of remote work from other confounding factors influencing real estate markets.

Parallel Trends Assumption

- Counties with a high share of workers working from home may exhibit different housing price trends even before 2020, potentially violating the parallel trends assumption required for Difference-in-Differences estimation.

Donor Pool Limitations in Synthetic Control

- Synthetic Control methods rely heavily on the availability of comparable donor counties. For technology-intensive counties such as San Francisco or Santa Clara, it may be difficult to identify suitable comparison counties with similar pre-treatment trends and economic structures.